

Question 1

1.1 A✓✓

1.2 C✓✓

1.3 B✓✓

1.4 D✓✓

1.5 B✓✓

1.6 A✓✓

1.7 D✓✓

1.8 A✓✓

1.9 C✓✓

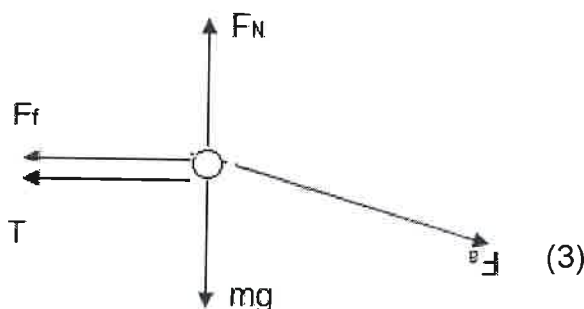
1.10 A✓✓

Question2

2.1 When a net force acts on a body, the body will accelerate in the direction of the force. This acceleration is directly proportional to the force and inversely proportional to the mass✓✓ (2)

2.2 $F_f = \mu N \checkmark = 0,15 \times 3(9,8) \checkmark = 4,41 \text{ N} \checkmark$ (3)

2.3



5 forces✓

directions✓

All correct arrows/touching ✓

2.4.1 $mg + 25 \sin 30 = N \checkmark$
 $1,5(9,8) + 25 \sin 30 \checkmark = 27,2 \text{ N}$
 $F_f = \mu N \checkmark = 0,15 \times 27,2 \checkmark = 4,08 \text{ N} \checkmark$ (4)

2.4.2 Block A: $F \cos \Theta - F_f - T = ma$
 $(1,5 \text{ kg}) \quad 25 \cos 30 - 4,08 - T = 1,5 a \checkmark$
 $17,57 - T = 1,5 a$

Block B: $T - F_f = ma$
 $(3 \text{ kg}) \quad T - 4,41 = 3 a \quad \checkmark$

Eq 1 + 2: $\checkmark \quad -1,5a + 17,57 - 4,41 = 3a$

$$13,16 = 4,5 a$$

$$a = 2,92 \text{ m.s}^{-2}$$

$T - 4,41 \checkmark = 3(2,92) = 13,18 \text{ N} \checkmark$ (5)

QUESTION 3

3.1 Only gravitational forces act on the ball ✓✓

(2)

3.2 $5,88 \text{ m.s}^{-1}$ ✓

(2)

3.3 $\Delta y = v_i \Delta t + \frac{1}{2} a t^2$ ✓

$$\Delta y = 5,88(2,6) + \frac{1}{2}(-9,8)(2,6)^2 \quad \checkmark \quad = -17,836 \text{ m}$$

17,836 m above the ground ✓

(3)

$$v_f^2 = v_i^2 + 2a\Delta y$$

$$(-19,5)^2 = 5,88^2 + 2(-9,8)\Delta y$$

3.4 $v_f = v_i + a\Delta t$

$$0 = 2,94 + (-9,8)t$$

$$t = 0,3 \text{ s} + 2,6 = 2,9 \text{ s}$$

Method ✓

Answer ✓

OR $\frac{3,2-2,6}{2} = 2,9 \text{ s}$

(2)

3.5 $\Delta y = v_i \Delta t + \frac{1}{2} a t^2$

$$\Delta y = 2,94(0,3) + \frac{1}{2}(-9,8)(0,3)^2$$

$$= 0,441 \text{ m}$$

OR

$$v_f^2 = v_i^2 + 2a\Delta y$$

$$0 = 2,94^2 + 2(-9,8)\Delta y$$

(3)

(3)

3.6 Hot air balloon

$$\Delta y = v_i \Delta t$$

$$\Delta y = 5,88(2,9)$$

$$= 17,052 \text{ m up}$$

values substituted ✓

$$y \text{ between ball and balloon: } 17,052 - 0,441 =$$

$$= 16,661 \text{ m away from each other}$$

(4)

QUESTION 4

4.1 Linear momentum in an isolated system is conserved✓✓ (2)

4.2 $\Sigma p_{\text{initial}} = \Sigma p_{\text{final}}$ ✓

$$1 \checkmark m_1(14) + 1200 \times 14 \checkmark \quad 25 m_1 + 1200(-15) = 14 m + (1200 \times 14)$$

$$m_1 = 3163,63 \text{ kg} \checkmark \text{ not overloaded} \checkmark \quad (5)$$

4.3 $F\Delta t = m(v_f - v_i)$ ✓

$$F\Delta t = 1200 \checkmark (14 - (-15)) \checkmark \quad | \quad 3163(14 - 25) \\ = 34\,800 \text{ Ns to the right} \checkmark \quad (4)$$

QUESTION 4

4.1 Linear momentum in an isolated system is conserved✓✓ (2)

4.2 $\Sigma p_{\text{initial}} = \Sigma p_{\text{final}}$ ✓
 $\checkmark m_1(14) + 1200 \times 14 \checkmark$
 $m_1 = 3163,63 \text{ kg} \checkmark \text{ not overloaded} \checkmark$ (5)

4.3 $F\Delta t = m(v_f - v_i) \checkmark$
 $F\Delta t = 1200 \checkmark (14 - (-15)) \checkmark$
 $= 34\,800 \text{ Ns to the right} \checkmark$ (4)

QUESTION 5

[12]

- 5.1 Mechanical energy of a system in free fall (only force acting is gravitational force) is conserved ✓✓ (2)

5.2 $(mgh + \frac{1}{2} mv^2)_{top} = (mgh + \frac{1}{2} mv^2)_{bottom}$ ✓

$$5(9,8)(5) + \frac{1}{2} 5 \times 0^2 = (0 + \frac{1}{2} 5v^2)_{bottom}$$
 ✓

$$v = 9,9 m \cdot s^{-1}$$
 ✓ (3)

- 5.3 No friction forces that acts on an object ✓, $F_{net} = 0$, Newtons first law v constant ✓ (2)

5.4 $W_{nc} = \Delta E_K + \Delta E_p$ ✓

$$F_f x \cos \theta = \left(\frac{1}{2} mv^2 - \frac{1}{2} mv^2 \right) + (mgh_f - mgh_i)$$

$$(18 \times x \times \cos 180) = \left(\frac{1}{2} 5 \times (0^2 - 9,9^2) \right) + 5 \times 9,8 \times (3 - 0)$$
 ✓

$$x = 5,44 m$$
 (5)

$$\sin \theta = \frac{3}{5,44} \checkmark = 33,47^\circ \checkmark$$

QUESTION 6

6.1 The frequency a person perceives when 2 objects are moving relative to each other in the same medium ✓✓ Source + listener have (2)

different velocities relative to the → not how it appears in the guideline doc.

6.2 $v = \lambda f$ ✓ $340 = 0,28f$ ✓ $f = 1214,29 \text{ Hz.}$ ✓ (3)

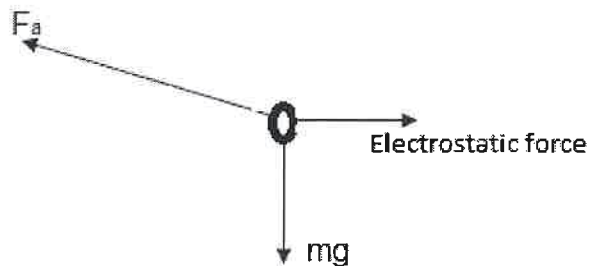
6.3 $F_L = \frac{v}{v - v_s} f_s$ ✓

$$F_L = \left(\frac{340}{340 - 30} \right) \times 1214,29 \checkmark$$

$$F_L = 1331,79 \text{ Hz} \checkmark \quad (5)$$

QUESTION 7

7.1.1



(3)

7.1.2 The magnitude of the electrostatic force exerted by one point charge (Q_1) on another point charge (Q_2) is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance (r) between them. ✓✓ (2)

7.1.3 $F = \frac{kQ_1Q_2}{r^2}$ ✓

$$F = \frac{9 \times 10^9 \times 5 \times 10^{-9} \times 9 \times 10^{-7}}{0,2^2} \checkmark$$

$$F = 10,125 \text{ N} \quad 0,110 \text{ N}$$

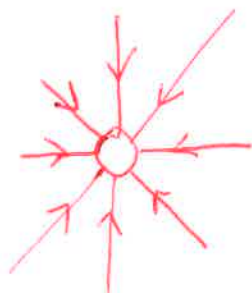
$$\sin 7 = \frac{10,125}{T} \checkmark$$

$$T = 83,08 \text{ N} \checkmark$$

$$\frac{0,1}{\sin 7} = 0,83 \text{ N}$$

(5)

7.2.1



(2)

7.2.2

$$E_{\text{net}} = \frac{kq}{r^2} \checkmark$$

$$\left(\frac{9 \times 10^9 \times 0,75 \times 10^{-6}}{0,09^2} \right) \checkmark$$

$$= 8,3 \times 10^5 \text{ N} \cdot \text{C}^{-1} \text{ to the left} \checkmark$$

(3)

QUESTION 8

8.1.1 $P = VI$ ✓

$$60 = 24 I$$
✓

$$I = 2,5 \text{ A}$$
✓

(3)

8.2 The ratio between voltage and current for a given resistor is constant at a constant temperature✓✓ *As in the document?* (2)

8.3 $V = IR$ ✓

8.1.3 $= 2,5 \times 7 = 17,5$ ✓✓

(3)

8.4 $V = 24 - 17,5 = 6,5 \text{ V}$ ✓
8.1.4

(2)

8.5 $I_{3\Omega} = \frac{V}{R} = \frac{6,5}{3} = 2,17 \text{ A}$ ✓
8.1.5

$$I_R = 2,5 - 2,17 = 0,33 \text{ A}$$

$$R = \frac{V}{I} = \frac{6,5}{0,33} = 19,7\Omega$$
✓

(5)

8.2.1 Next page

8.2.2 $m = \frac{\Delta y}{\Delta x} = \frac{9,15 - 11,75}{1,15 - 0,15} = -2,6$ ✓✓

8.2.3 $EMF = IR + Ir$

$$IR (Pd) = -Ir + EMF$$

$$y = mx + c$$

$$R = -\text{gradient} = 2,6 \Omega$$

8.2.4 12 V